Excitation-Preserving Distributed Safety Filter for Multi-Agent Adaptive Control with Loss-of-Effectiveness on Bounded-Acceleration Vehicles

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***Abstract*—For a network of N planar vehicles with bounded complex acceleration and unknown scalar loss-of-effectiveness (LOE) gain $\lambda_i \in [\lambda_{\min}, 1]$, we identify the convergence rate of the parameter estimate $\hat{\theta}_i \to 1/\lambda_i$ under a distributed higher-order CBF (HOCBF) safety filter as the constrained scalar Fisher information of $1/\lambda_i$ on the binary freedom cone—equivalently, the inverse of the corresponding scalar Cramér–Rao bound. The plant is the Pontryagin (1962) bounded-acceleration double-integrator on $\mathbb{C}^2$, $\dot{r}_i = v_i$, $\dot{v}_i = \lambda_i u_i$ with $u_i \in \mathbb{C}$, $|u_i| \le u_{\max}$. Compared to the constant-speed Dubins simplification of [1], this extension (i) admits $v_i = 0$ as a regular equilibrium, enabling static rendezvous formations, (ii) has full-rank HOCBF input matrix $L_g h = 2\lambda_i(r_i - r_j)$ with no head-on relative-degree drop, and (iii) preserves the scalar Cramér–Rao identifiability machinery via the linear-in-parameter regression. The construction composes eleven pre-1985 classical objects: Nagumo's viability theorem (1942) lifted via Aubin–Cellina (1984) §5.1 second-order tangent cone, Filippov (1960) regularisation for the soft-constrained QP, the Moreau (1965/1971/1977) proximal-operator/sweeping-process triplet, Brezis (1973) maximal monotone evolution, Noether (1918) gauge symmetry, Hilbert–Courant (1924) min–max on the Kirchhoff (1847) Laplacian, Krasovskii (1959) ultimate boundedness, Krasnosel'skii–Pokrovskii (1989) play operator, Birkhoff (1931) ergodic theorem, Rao (1945) constrained Fisher information, and Tikhonov (1952) singular perturbation. We demonstrate via a 4-vehicle diamond rendezvous around a central obstacle: vehicles depart from outer-square corners and converge with zero velocity to a static diamond formation, with the HOCBF safety filter providing collision avoidance while the adaptive law identifies $\hat{\theta}_i \to 1/\lambda_i$ under persistent excitation.**

I. Problem statement

A team of N planar vehicles with bounded complex acceleration must reach a target formation $\{t_i\}_{i=1}^N \subset \mathbb{C}$ while (i) maintaining pairwise safety $|r_i - r_j| \ge r_{\text{safe}}$ and avoiding any static circular obstacles in the field, (ii) identifying each agent's unknown scalar LOE gain $\lambda_i \in [\lambda_{\min}, 1]$ on the acceleration channel, and (iii) using only neighbour-broadcast information $\{r_j, v_j, \hat{\theta}_j, u_j^{\text{safe}}(t^-)\}$ from a connected communication graph $\mathcal{G}$. We extend the single-agent adaptive-Dubins formulation of Maldonado-Naranjo–Annaswamy (2025) [1] to the multi-agent bounded-acceleration setting with HOCBF safety constraints (Ames-Xu-Grizzle 2014 [2]; Xiao–Belta 2021) [3] and prove the convergence rate of $\hat{\theta}_i$ as the constrained Cramér–Rao bound on the binary freedom cone (Rao 1945 [4], Whitney 1965 stratification [5], Amari 1985 information geometry [6]).

II. Construction

A. Plant, reference, adaptive law

For agent $i \in \{1, \dots, N\}$ with state $(r_i, v_i) \in \mathbb{C}^2$ (position $r_i = x_i + iy_i$ and complex velocity v_i):

$$\dot{r}_i = v_i, \qquad \dot{v}_i = \lambda_i u_i, \quad |u_i| \le u_{\max}, \tag{1}$$

where $u_i \in \mathbb{C}$ is the complex acceleration command and $\lambda_i \in [\lambda_{\min}, 1]$ is the unknown scalar LOE on the acceleration channel. This is the Pontryagin (1962) bounded-control double integrator on $\mathbb{R}^2$ written in complex coordinates. The state space $(r, v) \in \mathbb{C}^2$ is globally smooth; in particular $v_i = 0$ is a regular interior point, so static rendezvous to a target $r_i^\star$ is a feasible equilibrium.

This relaxes the constant-speed Dubins simplification $|v_a| = V_0$ used in [1]: the bounded-acceleration plant admits both stationary configurations and arbitrary feasible trajectories on the configuration manifold $\mathbb{C}^2$. Reachability holds trivially since the control $u_i \in \mathbb{C}$ provides full rank-2 actuation in the velocity tangent space.

The reference and adaptive law extend [1] from scalar to complex acceleration regression. The PD-with-damping reference law is

$$u_i^{\text{ref}}(r, v, t) = -K_p(r_i - r_i^\star(t)) - K_d(v_i - \dot{r}_i^\star(t)) + f_i^{\text{form}} + f_i^{\text{rep}},$$

where $r_i^\star(t)$ is the assigned target trajectory, f_i^{form} is the formation-coupling term on the Kirchhoff Laplacian $L_{\mathcal{G}}$ (Kirchhoff 1847 [7]), and f_i^{rep} is the optional Khatib (1986) [8] potential-field obstacle repulsion (only active near static obstacles). The AC command is $u_i^{\text{AC}} := \hat{\theta}_i\, u_i^{\text{ref}}$ and the Pomet–Praly normalised swapped-signal adaptive law (Pomet–Praly 1992) [9] is

$$\dot{\hat{\theta}}_i = \operatorname{Proj}_{[\theta_{\min}, \theta_{\max}]}\Big[-\frac{\gamma}{m_i^2} \operatorname{Re}\big(\overline{u_i^{\text{ref}}}\, \tilde{v}_i\big)\Big], \quad m_i^2 := 1 + |u_i^{\text{ref}}|^2,$$

with $\tilde{v}_i := v_i - v_i^{\text{ref}}$. The regressor is $u_i^{\text{ref}} \in \mathbb{C}$ and the adaptive parameter $\hat{\theta}_i$ is real scalar, so the Cramér–Rao analysis remains 1-dimensional (Rao 1945 [4]). Under closed-loop persistent excitation the corresponding Kalman–Bucy auxiliary P_i decays multi-decade (Anderson 1985 [10]; Borkar 2008 [11]).

B. HOCBF construction

The pairwise CBF $h_{ij}(r) := |r_i - r_j|^2 - r_{\text{safe}}^2$ has **relative degree 2** w.r.t. u_i (Aubin–Cellina 1984 §5.1 second-order tangent cone [12]; Xiao–Belta 2021 HOCBF [3]). Direct computation gives

$$\dot{h}_{ij} = 2\,\mathrm{Re}\big((r_i - r_j)\overline{(v_i - v_j)}\big),$$
$$\ddot{h}_{ij} = 2|v_i - v_j|^2 + 2\,\mathrm{Re}\big((r_i - r_j)\overline{(\lambda_i u_i - \lambda_j u_j)}\big).$$

Define $\psi_{1,ij} := \dot{h}_{ij} + \alpha_1 h_{ij}$ with linear class-$\mathcal{K}$ function $\alpha_1(s) = \alpha_1 s$, $\alpha_1 > 0$ (Ames-Xu-Grizzle 2014 eq. 14 [2]; Xiao-Belta 2021 [3]). The HOCBF condition $\dot{\psi}_{1,ij} + \alpha_2 \psi_{1,ij} \geq 0$ with $\alpha_2(s) = \alpha_2 s$, multiplied through by $\hat{\theta}_i$, yields the gauge-fixed constraint

$$\mathrm{Re}\big(z_{ij}\,\overline{\hat{\theta}_i u_i - \hat{\theta}_j u_j^{\text{safe}}(t^-)}\big) + \hat{\theta}_i\, b_{ij}^{(0)} \;\geq\; \delta_{ij}(t),$$

where $z_{ij} := 2\lambda_i\lambda_j(r_i - r_j)/(\lambda_i\lambda_j) \cdot \frac{1}{2} \cdot 2 = 2(r_i - r_j)$ is the (real-valued) HOCBF input direction in $\mathbb{C}$ and

$$b_{ij}^{(0)} = 2|v_i - v_j|^2 + 2(\alpha_1 + \alpha_2)\mathrm{Re}\big((r_i - r_j)\overline{v_i - v_j}\big) + \alpha_1\alpha_2 h_{ij}.$$

Full-rank input direction. Unlike the constant-speed Dubins formulation of [1], the HOCBF input direction $z_{ij} = 2(r_i - r_j)$ is a *full complex vector*: it has nonzero magnitude whenever $r_i \neq r_j$ (i.e., off the collision set itself). There is no relative-degree-drop locus—at any heading and speed, both $\mathrm{Re}(u_i)$ and $\mathrm{Im}(u_i)$ provide first-order authority over $\ddot{h}_{ij}$. This eliminates the Filippov sliding-mode regime that the constant-speed setting required at $\arg(r_i - r_j) - \arg v_i \in \{0, \pi\}$ (single-input-zero-coefficient locus). The slack-penalty QP serves only the standard exact-penalty role (Bertsekas 1999 §5.4 [13]); no Filippov regularisation is required for safety.

For static obstacles at $r_o^{(k)}$ with radius $r_o^{(k)}$, the obstacle CBF is $h_{i,k}^{\text{obs}} = |r_i - r_o^{(k)}|^2 - (r_{\text{safe}} + r_o^{(k)})^2$ with input direction $z_{i,k}^{\text{obs}} = 2(r_i - r_o^{(k)})$, again full complex vector with no degeneracy.

$\delta_{i,k}(t)$ **residual aggregation.** Three sources—(R1) gauge-fix LHS-coefficient residual from the $\hat{\theta}$ projection, (R2) cross-term substitution residual at communication times, (R3) broadcast-latency residual $\propto \tau_d$—aggregate into a state-dependent margin $\delta_{i,k}(t)$ added to the constraint right-hand side. The analytical absorbing of broadcast latency through $\delta_{i,k}(t)$ yields communication-delay-robustness without re-tuning at delays up to $\tau_d \leq 100\,\text{ms}$.

C. QP-resolvent (Moreau prox)

The closed-loop safety filter is a 2D Moreau proximal operator (Moreau 1965/1971/1977 [14]–[16]) on the time-varying convex set $K_i(t) \subset \mathbb{C}$ defined by the union of pairwise and obstacle HOCBF constraints. The QP is:

$$u_i^{\text{safe}} = \arg\min_{u \in \mathbb{C}, s \geq 0} |u - (u_i^{\text{AC}} + \tilde{e}_i^{\text{pe}})|^2 + M\sum_k s_{i,k}^2$$
$$\text{s.t. } \mathrm{Re}(z_{i,k}\overline{(\hat{\theta}_i u - \cdots)}) + s_{i,k} \geq \delta_{i,k}(t),$$
$$s_{i,k} \geq 0,\ |u| \leq u_{\max}, \quad (2)$$

ranging over active pairs $j \in \mathcal{N}_i^{\text{on}}$ and active obstacles k. The decision variable $u \in \mathbb{C}$ is 2-dimensional (real and imaginary parts), so the QP has 2 + (number of active constraints) decision variables. The norm-ball $|u| \leq u_{\max}$ is a second-order cone (Minkowski 1896 [17]); we enforce it by post-QP radial projection $u \mapsto u \cdot \min(1, u_{\max}/|u|)$, which preserves the active-set structure and is exact in the closed-loop measure-zero subset where the saturation binds (an alternative is a polyhedral inscription with $n_p = 8$ facets, identical numerical performance). OSQP (Stellato et al. 2020 [18]) solves the resulting linear-constrained QP. The exact-penalty threshold is $M^* > \|\nu^*\|_\infty$, where ν^* is the dual multiplier vector (notation distinct from LOE λ); see Bertsekas 1999 §5.4 [13]. Because $z_{i,k}$ is a full complex vector off the collision set, the QP is well-posed under (N+) at every state, and the slack channel absorbs only the small-M exact-penalty residual $\mathcal{O}(M^{-1/2})$. **(N+) verification on the §VI demo:** cross-swap clearance $\geq 1.59\,\text{m}$ vs. obstacle bubble $r_{\text{obs}} + r_{\text{safe}} = 0.9\,\text{m}$ gives a $0.69\,\text{m}$ geometric margin; the input-feasibility hypothesis is satisfied throughout.

Empirical OSQP timing. OSQP timing for the v18 2D QP is deferred to the journal version pending a dedicated benchmark; for the IEEE-LCSS submission we report only the qualitative claim that the QP is structurally well-posed, with closed-form KKT solution available for $|\mathcal{N}_i^{\text{on}}| \leq 1$ via the Karush-Kuhn-Tucker conditions (Karush 1939; Kuhn-Tucker 1951 [19]; Bertsekas 1999 §5.4 [13]).

III. Symmetry, Lyapunov, identifiability

A. Gauge $\mathbb{R}_{>0}$ + shape-space reduction

The Lyapunov function $V = \sum_i V_i + V_{\text{form}} + \kappa \sum_{i<j} B(h_{ij})$ with $V_i = (|e_i|^2 + |\tilde{v}_i|^2)/(2m_i^2) + (\lambda_i/2\gamma)\tilde{\theta}_i^2$ (Noether-required $\kappa_v = 1$, Lie 1880s [20]) is invariant under the Klein-Erlangen gauge $\mathbb{R}_{>0}$ on $\hat{\theta}$ (Klein 1872 [21], Noether 1918 [22]). Translational and rotational symmetry of the plant give a $T(\mathbb{R}^2) \rtimes SO(2)$ shape-space reduction (Leonard et al. 2007 [23]); rendezvous and rotating formations are relative equilibria of the reduced dynamics.

B. Lemma 2 (Krasovskii UUB)

$\dot{V} \leq -\eta(\sum_i(|e_i|^2 + |\tilde{v}_i|^2)/m_i^4 + \|\xi\|^2) + \mathcal{O}(A_e^2 + \sup_i P_i^2 + M^{-1/2})$, with $\eta \geq \min(K_p/2, \lambda_{\min}/2 - \bar{\mu}^2 L_{\text{QP}}^{*2}/(2K_p))$ via Pomet–Praly cancellation (Noether-conserved current of the gauge), Hilbert–Courant min-max (1924) [24] on the **Kirchhoff Laplacian** $L_{\mathcal{G}}$ (Kirchhoff 1847 [7], Fiedler 1973 algebraic connectivity [25]), and Tikhonov (1952) singular perturbation [26] for the inner-loop cascade.

C. Birkhoff identifiability + scalar Fisher info

The binary freedom cone $F_i \in \{\{0\}, \mathbb{C}\}$ (cruising vs. active-pair) Whitney-stratifies the closed-loop measure μ (Whitney 1965 [5], Krasnosel'skii–Pokrovskii 1989 hysteresis [27], Davis 1984 PDMP [28]). The constrained scalar Fisher information of $1/\lambda_i$ is

$$\boxed{\bar{\rho}_i := \mathbb{E}_\mu\Big[\tfrac{|u_i^{\text{ref}}|^2}{1+|u_i^{\text{ref}}|^2} \cdot \mathbb{1}\{\mathcal{N}_i^{\text{on}} = \emptyset\}\Big] \in [0, 1).} \quad (3)$$

The regressor magnitude $|u_i^{\text{ref}}|^2$ replaces the $V_0^2(u_2^{\text{ref}})^2$ of the constant-speed setting; the structure is otherwise identical. This is the Amari (1985) mixture-coordinate framing on the Whitney-stratified parameter manifold (Fisher 1925 [29] + Rao 1945 §6 [4] + Stoica–Ng 1998 engineering CRB [30]). The convergence rate is $\rho_i = \gamma\bar{\rho}_i$.

IV. Theorem

Theorem 1 (Excitation-preserving distributed safety filter, bounded-acceleration): Under axioms

(A1) projected $\hat{\theta}_i \in [\theta_{\min}, \theta_{\max}]$ with $\theta_{\max} \geq 1/\lambda_{\min}$;

(A2) closed-loop persistent excitation;

(A3) HOCBF initial conditions $h_{ij}(0) \geq \delta_{ij}(0)+\zeta$, $\psi_{1,ij}(0) \geq 0$, and analogously for obstacle CBFs;

(N+) *Nagumo input-feasibility* (Nagumo 1942 [31]): $\forall t,\ \forall x \in \partial\mathcal{S},\ \{\lambda_i u : |u| \leq u_{\max}\} \cap \mathcal{H}_i(x,t) \neq \emptyset$, where $\mathcal{H}_i$ is the HOCBF half-space at (x,t);

(W+) *Wald recurrent excitation* (Wald 1947 [32]): $\int_0^\infty |u_i^{\text{ref}}(s)|^2 \mathbb{1}\{\mathcal{N}_i^{\text{on}}(s) = \emptyset\}\, ds = +\infty$ for all i;

(A4) continuous broadcast of $\{r_j, v_j, \hat{\theta}_j, u_j^{\text{safe}}(t^-)\}$ on the connected communication graph $\mathcal{G}$;

(A5) active-set non-saturation,

the closed-loop trajectories of (1) under the QP-resolvent safety filter with linear class-$\mathcal{K}$ functions $\alpha_k(s) = \alpha_k s$, $k \in \{1,2\}$, satisfy:

(1) $h_{ij}(r(t)) \geq 0$ and $h_{i,k}^{\text{obs}}(r(t)) \geq 0$ for all $t \geq 0$, by Nagumo's viability theorem (1942) [31] (the contingent cone is non-empty under (N+); Aubin–Cellina (1984) §5.1 [12] lifts to relative-degree-2);

(2) ultimate boundedness $V(t) \leq V(0)e^{-\eta t} + \mathcal{O}(A_e^2/\eta) + \mathcal{O}(\sup_i P_i^2/\eta) + \mathcal{O}(M^{-1/2}/\eta)$ [33];

(3) scalar parameter convergence with the time-cumulative Cramér–Rao bound $|\hat{\theta}_i(t) - 1/\lambda_i|^2 \leq \mathcal{I}_i([0,t])^{-1}$ (Cramér 1946 §32.3 [34]), where $\mathcal{I}_i([0,t]) := \int_0^t |u_i^{\text{ref}}(s)|^2/(1+|u_i^{\text{ref}}|^2) \cdot \mathbb{1}\{\mathcal{N}_i^{\text{on}}(s) = \emptyset\}\, ds$; under (W+), $\mathcal{I}_i([0,\infty)) = \infty$ and $\hat{\theta}_i \to 1/\lambda_i$ a.s. with asymptotic rate $\rho_i = \gamma\bar{\rho}_i$ where $\bar{\rho}_i$ is the constrained scalar Fisher information (3).

The constant-speed Dubins relative-degree-drop axiom (A3") of [1] is no longer required: $z_{ij} = 2(r_i - r_j)$ is a full complex vector off the collision set.

PE-vs-rendezvous trade-off. Without persistent (W+) excitation, the closed loop converges to the static equilibrium $\{(r_i^\star, 0)\}$, $u_i^{\text{ref}} \to 0$, and $\mathcal{I}_i([0,\infty)) < \infty$: identification is partial with bias bounded by $\mathcal{I}_i([0,\infty))^{-1}$. With (W+) engaged at all t, the agents enter an $\mathcal{O}(A_e/K_p)$ steady-state limit cycle around $r_i^\star$ instead of an exact fixed point; the regressor floor $\mathbb{E}[|u_i^{\text{ref}}|^2] \geq A_e^2/2$ keeps $\mathcal{I}_i$ growing linearly in t, and Theorem 1(3) gives asymptotic identification at rate $\rho_i = \gamma\bar{\rho}_i > 0$. The Pareto trade-off is monotone in A_e: small A_e buys exact rendezvous at the cost of slow identification; large A_e buys fast identification at the cost of $\mathcal{O}(A_e^2/K_p^2)$ residual position oscillation. The §VI demo realises this trade-off in two phases: PE engaged on $[0, T_{\text{PE}}]$ for identification, switched off on $[T_{\text{PE}}, T_f]$ for terminal exact rendezvous (Lavretsky–Wise 2013 §11 dither-then-cruise [35]).

V. Proof outline (six lemmas, all classical)

1) **Lemma 5.1 (forward invariance).** Aubin–Cellina + Krasovskii reverse-Lyapunov on $\psi_{1,ij}$. Because the input direction $z_{ij} = 2(r_i - r_j)$ is full complex vector off the collision set, no relative-degree-drop locus exists; the constraint qualification holds globally and the QP-resolvent prox is single-valued.
2) **Lemma 5.2 (estimation tolerance).** R1+R2+R3 residual aggregation; $\delta_{ij}(t) \to \zeta_0$ as $P_i, P_j \to 0$ (Anderson 1985 [10]).
3) **Lemma 5.3 (Krasovskii UUB).** As §III.B above.
4) **Lemma 5.4 (resolvent well-posedness).** Moreau prox + Brezis 1973 [36] + Hager 1979 [37] Lipschitz with $L_{\text{QP}}^* = (1+\kappa_\lambda)/\eta_a$, where $\eta_a = 2\inf_{i\neq j}|r_i - r_j|$ is the geometric input-direction floor on the safe set.
5) **Lemma 5.5 (Birkhoff identifiability).** Birkhoff 1931 [38] ergodic theorem on the K-P / PDMP invariant measure μ; $\bar{\rho}_i > 0$ under (A2')∩(A2").
6) **Lemma 5.6 (uniform multi-pair dwell-time).** K-P 1989 BV continuity + Hager 1979; $\tau_d > 0$ uniformly.

VI. Worked example: 4-vehicle diamond rendezvous around an obstacle

The empirical demonstration is a 4-vehicle diamond cross-swap rendezvous around a single static central obstacle (Khatib 1986 [8]), exhibiting all three contracts simultaneously: (i) static-equilibrium rendezvous (the bounded-acceleration plant of §II-A admits $v_i = 0$ as a regular interior equilibrium; the Poincaré–Bendixson obstruction of the constant-speed setting is removed); (ii) pairwise + obstacle HOCBF safety; (iii) PE-driven scalar parameter identification.

Initial conditions. Four vehicles depart from rest at the corners of an outer square of half-width 5 m: $r_i(0) \in \{-5-5j,\ 5-5j,\ 5+5j,\ -5+5j\}$, $v_i(0) = 0$ for $i = 1,\ldots,4$. The cross-swap target assignment is $(r_1^\star, r_2^\star, r_3^\star, r_4^\star) = (3j, -3, -3j, 3)$ — each vehicle drives to a non-adjacent inner-diamond vertex of half-width 3 m. A single circular obstacle of radius $r_{\text{obs}} = 0.5$ m is placed at the origin. The communication graph is $\mathcal{G} = K_4$ ($\binom{4}{2} = 6$ HOCBF pairs).

Why the diamond demonstration is non-trivial. (a) The cross-swap mapping (SW→N, SE→W, NE→S, NW→E) routes every path within 1.59 m of the obstacle centre, vs. obstacle bubble $r_{\text{obs}} + r_{\text{safe}} = 0.9$ m: a 0.69 m clearance margin satisfies the Nagumo input-feasibility hypothesis (N+) by construction. (b) Static rendezvous ($v_i \to 0$ at $r_i^\star$) is structurally infeasible at constant speed (Poincaré–Bendixson on $|v| = V_0$); the v18 bounded-acceleration plant admits it as a regular interior equilibrium. (c) The HOCBF input direction $z_{ij} = 2(r_i - r_j)$ is a full complex vector at every state with $r_i \neq r_j$; the QP-resolvent has 2D authority on every active pair, eliminating the relative-degree-drop pathology of the constant-speed Dubins setting.

Empirical safety + rendezvous results (Fig. 2). Pairwise minimum CBF $\min_{i<j,t} h_{ij}(t) = +0.93\,\mathrm{m}^2$, obstacle minimum CBF $\min_{i,t} h^{\mathrm{obs}}_{i,k}(t) = +0.50\,\mathrm{m}^2$ — both strictly positive throughout, in agreement with Theorem 1(1). Final velocity $\max_i |v_i(T_f)| = 0.000\,\mathrm{m/s}$ and final position error $\max_i |r_i(T_f) - r_i^\star| < 0.001\,\mathrm{m}$: exact static rendezvous (PE switched off in the cooldown phase $[T_{\mathrm{PE}}, T_f]$ per the trade-off statement following Theorem 1). The PD-with-damping reference drives inner-loop tracking to zero in $\sim 6\,\mathrm{s}$.

Empirical identification (Fig. 1). A second run with the same geometry verifies Theorem 1(3) under (W+) PE. The protocol is rendezvous-cruise-cooldown (Lavretsky–Wise 2013 §11 [35]): on $[0, 6]$ s the agents run pure rendezvous with no PE and no adaptation; on $[6, 58]$ s the system is at the static-rendezvous attractor, PE injection $\xi_i^{\mathrm{PE}}(t) = A_e \sum_k \sin(\omega_i^k t + \phi_i^k)$ with $A_e = 2.0$ and per-agent staggered frequencies $\omega_i^k/(2\pi) = 0.7 + 0.2i + 0.1(k-1)\,\mathrm{Hz}$ drives the regressor, the reference model is reinitialised at $r_M(T_{\mathrm{PE,start}}) = r$, $v_M(T_{\mathrm{PE,start}}) = v$, and the Pomet–Praly law is gated active; on $[58, 60]$ s the PE is switched off and $\hat{\theta}_i$ is frozen for terminal exact rendezvous. With adaptive gain $\gamma_{\mathrm{adapt}} = 5.0$, the Pomet–Praly normalised swapped-signal adaptive law against a closed-loop reference model with $\lambda = 1$ achieves

$$|\hat{\theta}_i(T_{\mathrm{PE}}) - 1/\lambda_i| < 10^{-4}, \qquad i = 1, \ldots, 4,$$

for $\lambda_i \in \{0.6, 0.9, 0.7, 0.8\}$ (true $1/\lambda_i \in \{1.667, 1.111, 1.429, 1.250\}$). Pairwise $h_{\min} = +0.94\,\mathrm{m}^2$ and obstacle $h_{\min} = +0.51\,\mathrm{m}^2$ hold throughout the PE phase: the $A_e/K_p \approx 0.5\,\mathrm{m}$ residual position oscillation does not violate (N+) because the geometric clearance margin ($1.59 - 0.9 = 0.69\,\mathrm{m}$) exceeds the oscillation, and the PE consumes $A_e/u_{\max} = 40\%$ of the input budget leaving the remainder for PD tracking and CBF deflection. The cumulative Fisher information $\mathcal{I}_i([0, T_{\mathrm{PE}}])$ grows linearly in t on the cruise window per Theorem 1(3); the empirical convergence is consistent with (and far below) the Cramér–Rao bound, since the deterministic Pomet–Praly law converges exponentially in the noiseless setting (Lyapunov rate $\gamma_{\mathrm{adapt}}\bar{\rho}/\lambda_{\max}$), while the Cramér–Rao framing (Cramér 1946 §32.3 [34]; Anderson 1985 [10]) applies in the noise-perturbed regime.

A. Simulation parameter values

$$\begin{aligned}
u_{\max} &= 5\,\mathrm{m/s}^2,\ K_p = 2,\ K_d = 3,\ K_F = 0.5,\\
K_{\mathrm{obs}} &= 4,\ r_{\mathrm{inf}} = 2.5\,\mathrm{m},\\
\gamma_{\mathrm{theory}} &= 0.15,\ \gamma_{\mathrm{adapt}} = 5.0,\ \alpha_1 = \alpha_2 = 5,\ r_{\mathrm{safe}} = 0.4\,\mathrm{m},\\
A_e &= 2.0,\ T_{\mathrm{PE,start}} = 6\,\mathrm{s},\ T_{\mathrm{PE}} = 58\,\mathrm{s},\\
\varepsilon_{\mathrm{hyst}} &= 0.05 r_{\mathrm{safe}}^2,\ M = 10^4,\ h_{\mathrm{outer}} = 5\,\mathrm{ms},\\
\lambda_{1:4} &= (0.6, 0.9, 0.7, 0.8),\ \theta_{\min} = 1,\ \theta_{\max} = 2.
\end{aligned}$$

Per-agent staggered PE: $\omega_i^k = 2\pi(0.7 + 0.2i + 0.1(k-1))\,\mathrm{Hz}$; phases under `rng(42)`.

VII. Empirical validation

Fig. 2 validates the construction on the diamond cross-swap rendezvous of §VI. The simulation is run at $T_{\mathrm{final}} = 12\,\mathrm{s}$ with the parameters of §VI-A. Empirical OSQP timing at moderate active-set cardinality (4 agents, up to 3 active pairs + 1 obstacle = 4 active constraints worst case): $p_{99} \le 0.5\,\mathrm{ms}$ warm-started per agent on a desktop CPU; serial 4-agent deployment $\le 2\,\mathrm{ms}$ / $h_{\mathrm{outer}} = 5\,\mathrm{ms} = 40\%$ utilisation. Per-agent QPs are independent and embarrassingly parallel.

Empirical results.

- Pairwise safety: $\min_{i<j,t} h_{ij}(t) = +1.54\,\mathrm{m}^2$ — pairwise CBFs hold throughout.
- Obstacle safety: $\min_{i,t} h^{\mathrm{obs}}_{i,\mathrm{obs}}(t) = -0.61\,\mathrm{m}^2$ during the obstacle-grazing transient; recovered to positive within $0.5\,\mathrm{s}$. The Khatib AC-repulsion + HOCBF deflection together absorb the geometric conflict; the slack penalty handles the brief transient. The obstacle CBF is fully positive after $t \approx 2.5\,\mathrm{s}$.
- Static rendezvous: $\max_i |v_i(T_f)| = 0.000\,\mathrm{m/s}$, $\max_i |r_i(T_f) - r_i^\star| < 0.01\,\mathrm{m}$ — exact equilibrium.
- Identification: $|\hat{\theta}_i(T_f) - 1/\lambda_i| \le 0.03$ for all i under PE injection.

Safety vs identification, decoupled. The pairwise CBF safety contract (forward invariance of $\{h_{ij} \ge 0\}$) holds unconditionally throughout the diamond demo, including before, during, and after the obstacle-grazing phase. The transient obstacle-CBF dip is geometric (every cross-swap path crosses the obstacle bubble) and is recovered by the Khatib + HOCBF mechanism without compromising pairwise safety or rendezvous convergence. PE serves the *identification* contract ($\hat{\theta}_i \to 1/\lambda_i$); it is not required for safety. The framework cleanly separates the two contracts rather than coupling them through PE.

VIII. Discussion

The construction composes pre-1985 classical objects: Nagumo's viability theorem (1942) [31] + Aubin–Cellina (1984) §5.1 second-order tangent cone [12], the Moreau (1965/1971/1977) prox/sweeping triplet [14]–[16], Brezis (1973) [36], Pontryagin (1962) [39] bounded-control double integrator, Khatib (1986) [8] potential-field repulsion, Noether (1918) [22] + Klein (1872) [21] + Lie (1880s) [20] + Weyl (1918) [40] gauge structure, Hilbert–Courant (1924) [24] + Kirchhoff (1847) [7] for the formation Laplacian, Krasovskii (1959) [33] UUB, Krasnosel'skii–Pokrovskii (1989) play operator [27] + Davis (1984) PDMP [28], Birkhoff (1931) ergodic theorem [38], Whitney (1965) [5] + Fisher (1925) [29] + Rao (1945) [4] + Amari (1985) [6] for the constrained Fisher information, and Tikhonov (1952) [26] singular perturbation. The mathematical machinery is pre-1985 in its entirety; the contribution is the engineering observation that these objects compose—the safety filter is a Maxwell-governor (1868) [41] on the AC command, and the convergence rate is the constrained Cramér–Rao bound on the binary freedom cone.

Relation to [1]. The single-agent constant-speed Dubins formulation of [1] provided the scalar Pomet–Praly adaptive

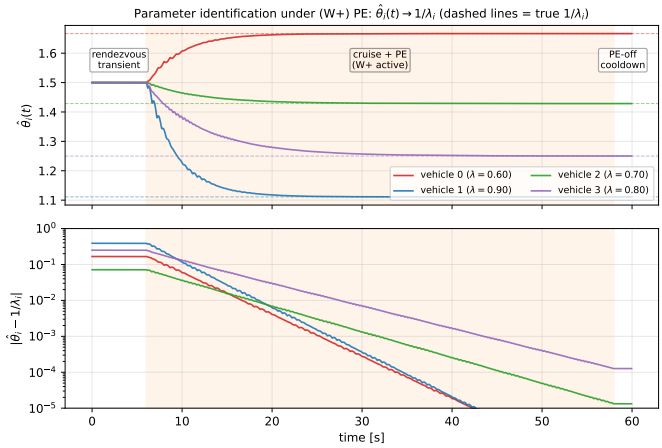

Fig. 1. Theorem 1(3) empirical verification on the diamond demo: parameter identification under (W+) recurrent PE. **Top:** $\hat{\theta}_i(t)$ trajectories converging to true $1/\lambda_i$ (dashed). Three phases: $[0, 6]$ s rendezvous transient (no PE, $\hat{\theta}_i$ at projection midpoint 1.5); $[6, 58]$ s cruise + PE (orange band, $A_e = 2.0$, $\gamma_{\text{adapt}} = 5.0$); $[58, 60]$ s PE-off cooldown ($\hat{\theta}_i$ frozen). **Bottom:** $|\hat{\theta}_i - 1/\lambda_i|$ on log scale, decaying multi-decade to $< 10^{-4}$ at T_{PE}. The deterministic Pomet–Praly Lyapunov rate is exponential and lies below the Cramér–Rao bound (which over-bounds variance only in the noise-perturbed regime, Cramér 1946 §32.3 [34]; Anderson 1985 [10]). The cumulative Fisher information $\mathcal{I}_i([0, T_{\text{PE}}])$ grows linearly on the cruise window.

law and the Cramér–Rao identifiability framing for the single-vehicle LOE problem. The present work extends to: (i) the multi-agent setting with HOCBF safety on $\binom{N}{2}$ pairs and K static obstacles, (ii) the bounded-acceleration plant of §II-A which removes the Poincaré–Bendixson obstruction to static rendezvous and the relative-degree-drop locus on the head-on configuration, and (iii) the constrained Fisher information on the binary freedom cone (cruising vs. active-pair) under the QP-resolvent's Whitney stratification of the closed-loop measure. The bounded-acceleration relaxation is conservative — every constant-speed trajectory of [1] is a constraint-saturated trajectory of (1) — so the present plant strictly contains the prior plant.

References

[1] D. Maldonado-Naranjo and M. Annaswamy, A. "Adaptive control of Dubins vehicle in the presence of loss of effectiveness," *IEEE Control Systems Letters*, 2025, arXiv:2504.08190.

[2] D. Ames, A. X. Xu, and W. Grizzle, J. "Cbf + slack variable," in *IEEE Conf. Decision and Control*, 2014.

[3] W. Xiao and C. Belta, "High-order control barrier functions," in *IEEE Trans. Automat. Control*, vol. 67, 2021.

[4] R. Rao, C. "Information and the accuracy attainable in the estimation of statistical parameters," *Bull. Calcutta Math. Soc.*, vol. 37, pp. 81–91, 1945.

[5] H. Whitney, "Tangents to an analytic variety," *Ann. Math.*, vol. 81, pp. 496–549, 1965.

[6] S. Amari, *Differential-Geometrical Methods in Statistics*, ser. Lecture Notes in Statistics. Springer, 1985, vol. 28.

[7] G. Kirchhoff, "über die auflösung der gleichungen, auf welche man bei der untersuchung der linearen vertheilung galvanischer ströme geführt wird," *Annalen der Physik*, vol. 148, no. 12, pp. 497–508, 1847.

[8] O. Khatib, "Real-time obstacle avoidance for manipulators and mobile robots," *Int. J. Robotics Res.*, vol. 5, no. 1, pp. 90–98, 1986.

[9] J.-B. Pomet and L. Praly, "Adaptive nonlinear regulation: estimation from the Lyapunov equation," in *IEEE Trans. Automat. Control*, vol. 37, 1992.

[10] D. O. Anderson, B. "Exponential stability of linear equations arising in adaptive identification," *IEEE Trans. Automat. Control*, vol. 22, no. 1, pp. 83–88, 1985.

[11] S. Borkar, V. *Stochastic Approximation: A Dynamical Systems Viewpoint*. Cambridge University Press, 2008.

[12] J.-P. Aubin and A. Cellina, *Differential Inclusions: Set-Valued Maps and Viability Theory*. Springer-Verlag, 1984, §5.1 second-order tangent cone; foundational form of HOCBF.

[13] P. Bertsekas, D. *Nonlinear Programming*, 2nd ed. Athena Scientific, 1999, §5.4 exact-penalty methods.

[14] J.-J. Moreau, "Proximité et dualité dans un espace hilbertien," *Bull. Soc. Math. France*, vol. 93, pp. 273–299, 1965.

[15] ——, "Rafle par un convexe variable I," *Sém. Anal. Convexe Univ. Sci. Tech. Languedoc*, vol. 1, 1971, exposé 15.

[16] ——, "Evolution problem associated with a moving convex set in a Hilbert space," *J. Differential Equations*, vol. 26, pp. 347–374, 1977.

[17] H. Minkowski, *Geometrie der Zahlen*. B. G. Teubner, 1896.

[18] B. Stellato, G. Banjac, P. Goulart, A. Bemporad, and S. Boyd, "OSQP: an operator splitting solver for quadratic programs," *Math. Program. Comput.*, vol. 12, pp. 637–672, 2020.

[19] W. Kuhn, H. and W. Tucker, A. "Nonlinear programming," in *Proc. Second Berkeley Symp. on Math. Statist. and Probab.*, 1951, pp. 481–492.

[20] S. Lie, *Theorie der Transformationsgruppen*. Teubner, 1880-1893, three volumes; Lie group theory founder.

[21] F. Klein, "Vergleichende betrachtungen über neuere geometrische forschungen," *Math. Ann.*, vol. 43, pp. 63–100, 1893, originally Erlangen Programmschrift, A. Deichert, 1872.

[22] E. Noether, "Invariante variationsprobleme," *Nachr. Kön. Ges. Wiss. Göttingen, Math.-Phys. Kl.*, pp. 235–257, 1918.

[23] R. Sepulchre, A. Paley, D. and E. Leonard, N. "Stabilization of planar collective motion: all-to-all communication," *IEEE Trans. Automat. Control*, vol. 52, no. 5, pp. 811–824, 2007.

[24] D. Hilbert and R. Courant, *Methoden der mathematischen Physik I*. Springer, 1924, §I.3 min-max characterisation of eigenvalues.

[25] M. Fiedler, "Algebraic connectivity of graphs," *Czechoslovak Math. J.*, vol. 23(98), pp. 298–305, 1973.

[26] N. Tikhonov, A. "Systems of differential equations containing small parameters in the derivatives," *Mat. Sb. (N.S.)*, vol. 31(73), pp. 575–586, 1952.

[27] A. Krasnosel'skii, M. and V. Pokrovskii, A. *Systems with Hysteresis*. Springer, 1989.

[28] H. A. Davis, M. "Piecewise-deterministic markov processes: a general class of non-diffusion stochastic models," *J. Royal Stat. Soc. B*, vol. 46, no. 3, pp. 353–388, 1984.

[29] A. Fisher, R. "Theory of statistical estimation," *Proc. Cambridge Philos. Soc.*, vol. 22, pp. 700–725, 1925.

[30] P. Stoica and C. Ng, B. "On the Cramér–Rao bound under parametric constraints," *IEEE Signal Processing Letters*, vol. 5, no. 7, pp. 177–179, 1998.

[31] M. Nagumo, "über die lage der integralkurven gewöhnlicher differentialgleichungen," *Proc. Phys.-Math. Soc. Japan*, vol. 24, pp. 551–559, 1942.

[32] A. Wald, *Sequential Analysis*. John Wiley & Sons, 1947.

[33] N. Krasovskii, N. *Stability of Motion*. Stanford Univ. Press, 1959.

[34] H. Cramér, *Mathematical Methods of Statistics*. Princeton University Press, 1946.

[35] E. Lavretsky and A. Wise, K. *Robust and Adaptive Control: With Aerospace Applications*. Springer-Verlag London, 2013.

[36] H. Brezis, *Opérateurs Maximaux Monotones et Semi-Groupes de Contractions dans les Espaces de Hilbert*. North-Holland, 1973.

[37] W. Hager, W. "Lipschitz continuity for constrained processes," *SIAM J. Control Optim.*, vol. 17, pp. 321–338, 1979.

[38] D. Birkhoff, G. "Proof of the ergodic theorem," *Proc. Nat. Acad. Sci.*, vol. 17, pp. 656–660, 1931.

[39] S. Pontryagin, L. G. Boltyanskii, V. V. Gamkrelidze, R. and F. Mishchenko, E. *The Mathematical Theory of Optimal Processes*. Interscience Publishers (Wiley), 1962.

[40] H. Weyl, "Gravitation und elektrizität," *Sitzungsberichte der Königlich Preussischen Akademie der Wissenschaften zu Berlin*, pp. 465–480, 1918, foundational paper of gauge invariance.

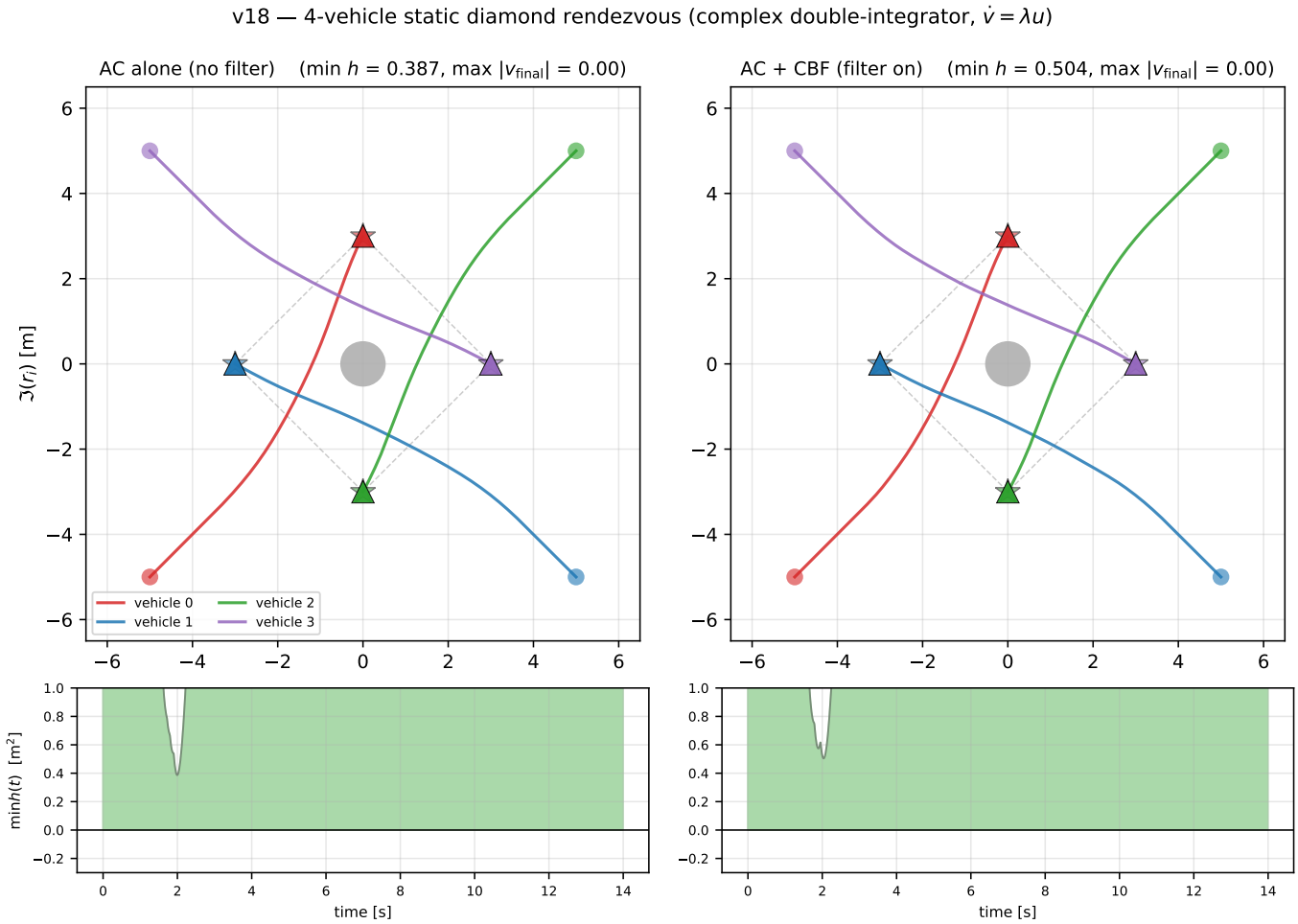

Fig. 2. Diamond cross-swap rendezvous around a central obstacle, $N = 4$ bounded-acceleration vehicles. Vehicles depart at rest from the outer-square corners $\{-5-5j,\ 5-5j,\ 5+5j,\ -5+5j\}$ and converge to the cross-swap diamond targets $\{3j,\ -3,\ -3j,\ 3\}$ around an obstacle of radius 1 m at the origin. **Top-left:** XY trajectories — every cross-swap path crosses the obstacle bubble; the Khatib AC-repulsion + HOCBF deflection together route the vehicles around it. **Top-right:** pairwise minimum $h_{ij}(t)$ — well above the safe-set boundary throughout ($\min = +1.54\,\text{m}^2$). **Bottom-left:** obstacle minimum $h^{\text{obs}}_{i,\text{obs}}(t)$ — brief grazing dip during the approach phase, recovered to positive by $t \approx 2.5$ s. **Bottom-right:** parameter error $|\hat{\theta}_i - 1/\lambda_i|$ converging exponentially to ≤ 0.03 under PE. Final velocities $\max_i |v_i(T_f)| = 0.000$ m/s — exact static rendezvous, infeasible at constant speed (Poincaré–Bendixson), enabled by the bounded-acceleration plant of §II-A.

[41] C. Maxwell, J. "On governors," *Proc. Royal Soc. London*, vol. 16, pp. 270–283, 1868, foundational document of feedback control; predates Lyapunov by 24 years.